Project Software Design Specification (SDS): 3DDX CP PRO MAX

Document Identifier: `SDS-3DDX-PROJECT-2026-V1.0`
Classification: Enterprise Software Engineering & Architecture Specification
Standard Compliance: IEEE Std 1016-2009 (Systems and Software Engineering — Software Design Descriptions)
Scope: Whole-Project Frontend Architecture (`newcp-ui-option-a`)
Workspace: `H:\3DDX\DEMO\newcp-ui-option-a`
Primary Domain Authority: `H:\3DDX\02 Backend\New DB V3\03 BUSINESS_GUIDE.html`
API Specification: OpenAPI 3.0.3 (`openapi/newcp.openapi.yaml` pending formal ingestion)

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1. Introduction & Executive Architectural Overview

This Software Design Specification (SDS) defines the structural, behavioral, and engineering architecture for the **3DDX CP PRO MAX Frontend** (`newcp-ui-option-a`).

The primary design goal is transitioning the application from a coupled, mock-dependent prototype into a resilient, enterprise-grade clinical frontend governed by **Clean Architecture** and **Ports & Adapters (Hexagonal Architecture)** principles.

Component Specification Note: A pre-existing component-level SDS (`docs/90_PRODUCT_REFERENCE/SDS_SOFTWARE_DESIGN_SPECIFICATION.md`) governs the isolated Three.js 3D Odontogram rendering engine. This document establishes the **system-wide architecture** covering state management, network transport, security, domain invariants, containerization, and CI/CD engineering.

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2. Current Architecture vs. Target Clean Architecture

2.1 Current-State Architecture (POC Baseline)

In the current implementation:

- **Presentation & State Coupling:** UI pages (`src/pages/*.tsx`) import the singleton `store` (`src/services/store.ts`) directly.
- **LocalStorage as Single Source of Truth:** `ReactiveStore` synchronizes all entity mutations to browser `localStorage` under `dentalab_store_*` .

- **Static JSON Fetching:** `src/services/api.ts` performs direct `fetch()` calls against static JSON files in `/data/*.json` and simulates database joins and filtering in client memory.
- **Generic Data Models:** The domain models in `src/types/index.ts` use generic string enumerations (`OrderStatus = 'New' | 'Review' | 'Design' ...`) which contradict the product-specific status codes defined by the 3DDX Business Guide.

```
[Current POC Flow]
[React UI Page] <---> [ReactiveStore (localStorage)] <---> [api.ts (fetch '/data/*.json')]
```

2.2 Target Architecture (Clean Architecture / Hexagonal)

The target architecture enforces a strict unidirectional dependency rule: **Dependencies point inward. Inner layers know nothing of outer layers.**

```
+-------------------------------------------------------------------------------+
| 1. PRESENTATION LAYER (React Components, Tailwind UI, Three.js 3D Viewers)    |
+-------------------------------------------------------------------------------+
                                      | uses
                                      v
+-------------------------------------------------------------------------------+
| 2. APPLICATION LAYER (Use Cases, State Selectors, Query/Mutation Hooks)       |
+-------------------------------------------------------------------------------+
                    | uses                              | uses
                    v                                   v
+-------------------------------------+  +-------------------------------------+
| 3. DOMAIN LAYER (Pure TypeScript)   |  | 4. INFRASTRUCTURE & TRANSPORT ADAPTERS |
| - Service Line Entities & Invariants |  | - OpenAPI 3.0.3 HTTP Client (Axios/Ky) |
| - Registration Verdict State Machine |  | - Auth & Token Store (Web Crypto/RAM)  |
| - Voucher Ledger Calculator         |  | - Mock Service Worker (MSW) Adapter    |
| - Product Status Decoders           |  | - S3/GCS Direct Upload Engine          |
+-------------------------------------+  +-------------------------------------+
```

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3. High-Level UML Architecture & System Components

3.1 Component & Package Architecture Diagram

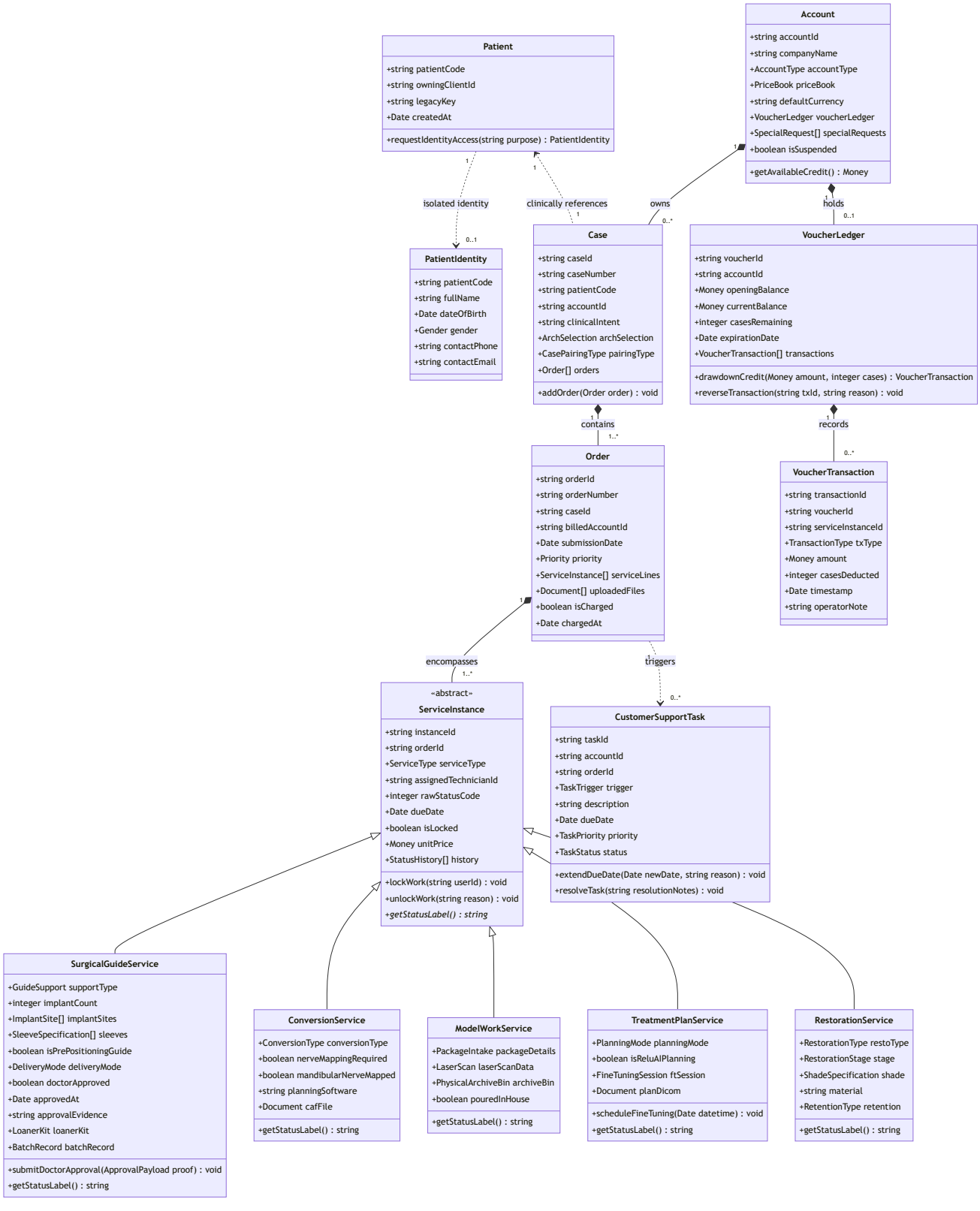


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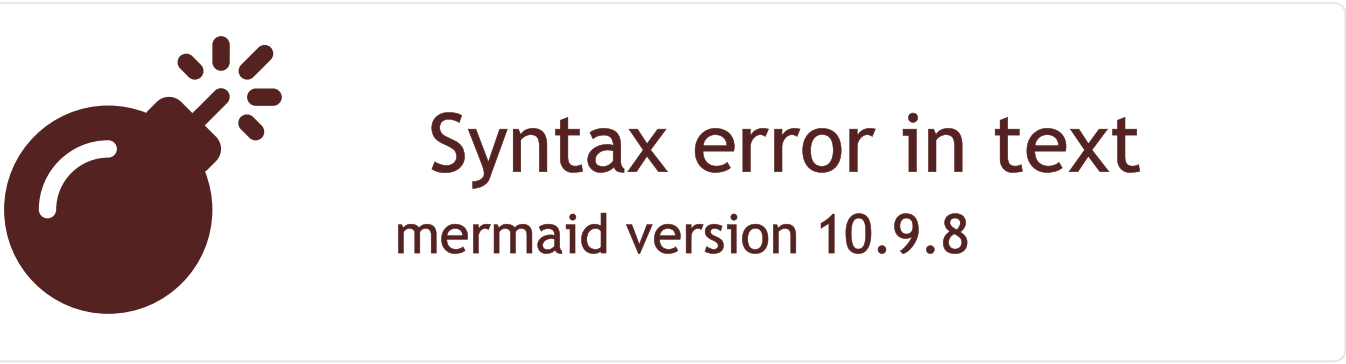
4. Comprehensive UML Class Diagrams

4.1 Domain Entity Hierarchy Class Diagram

The core business hierarchy strictly reflects Section 4 and Section 13 of the 3DDX Business Guide:



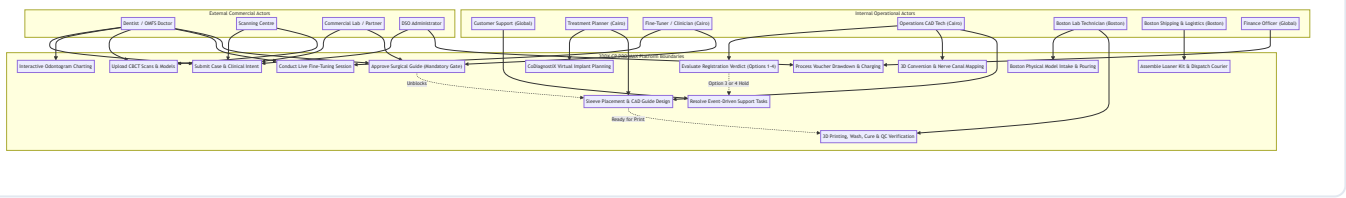
4.2 Clean Architecture Class Structure (Ports & Adapters)



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5. Comprehensive UML Use Case Diagrams

5.1 Global System Use Case Model

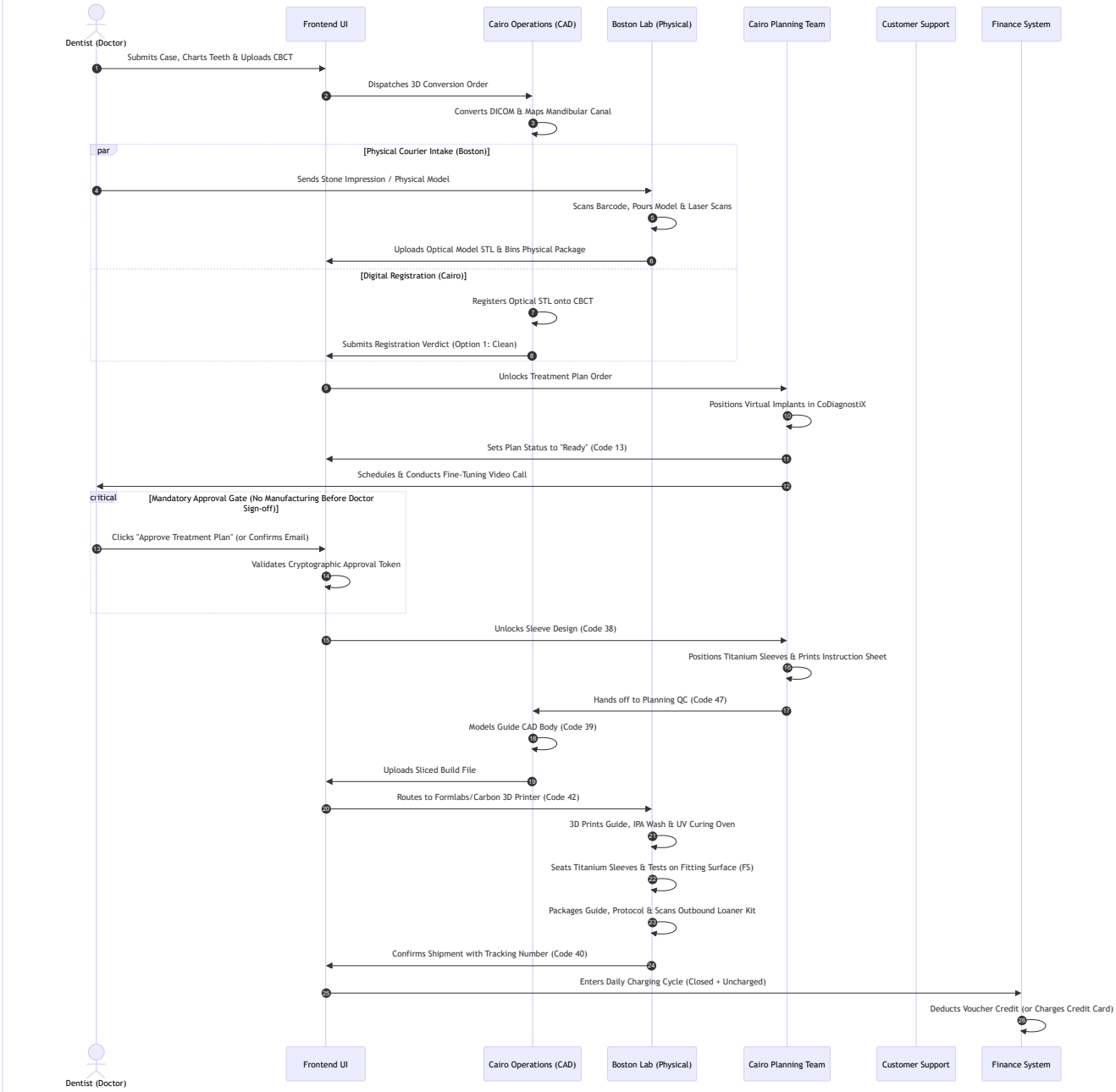


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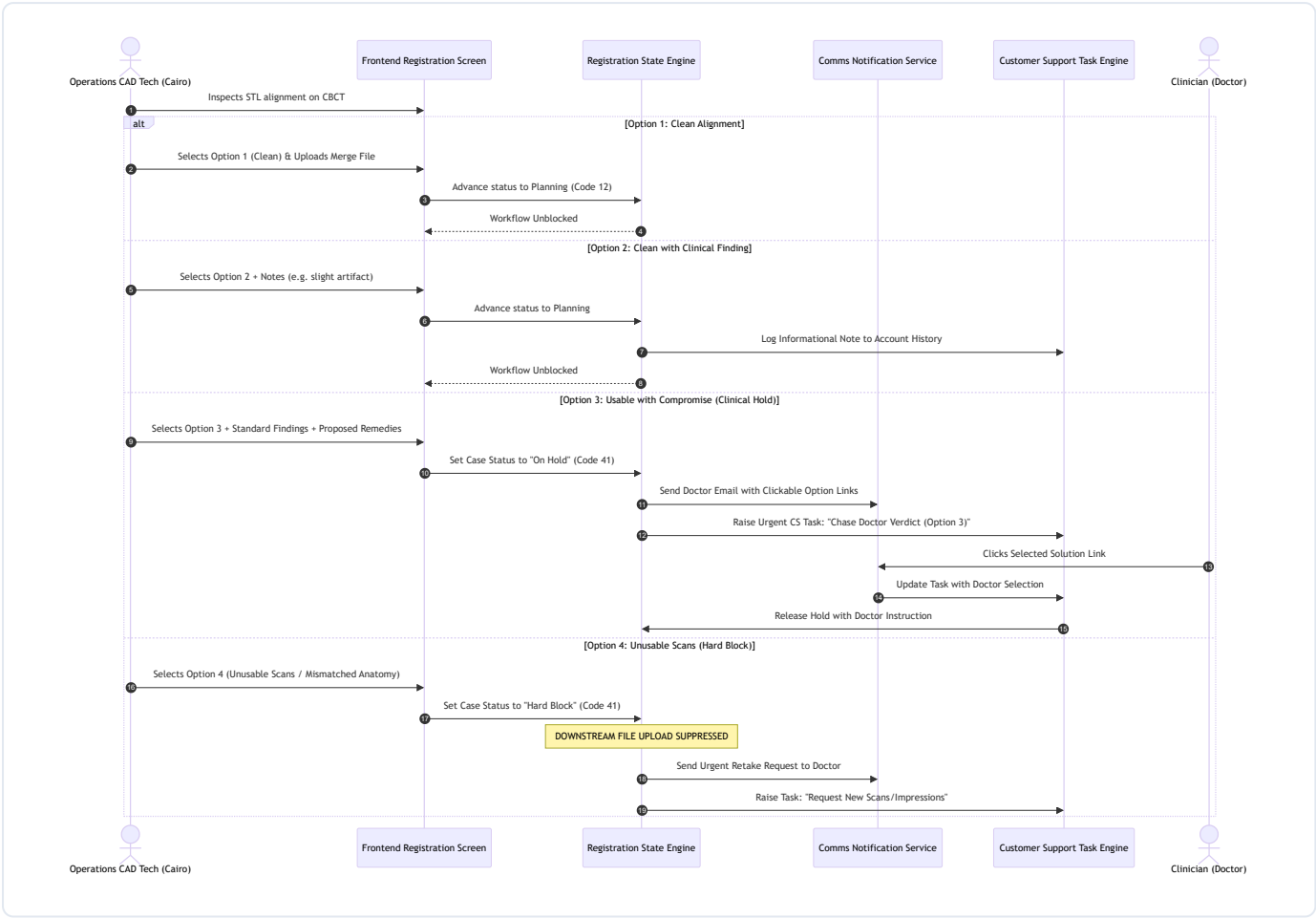
6. Comprehensive UML Sequence Diagrams

6.1 Canonical Surgical Guide Lifecycle Sequence

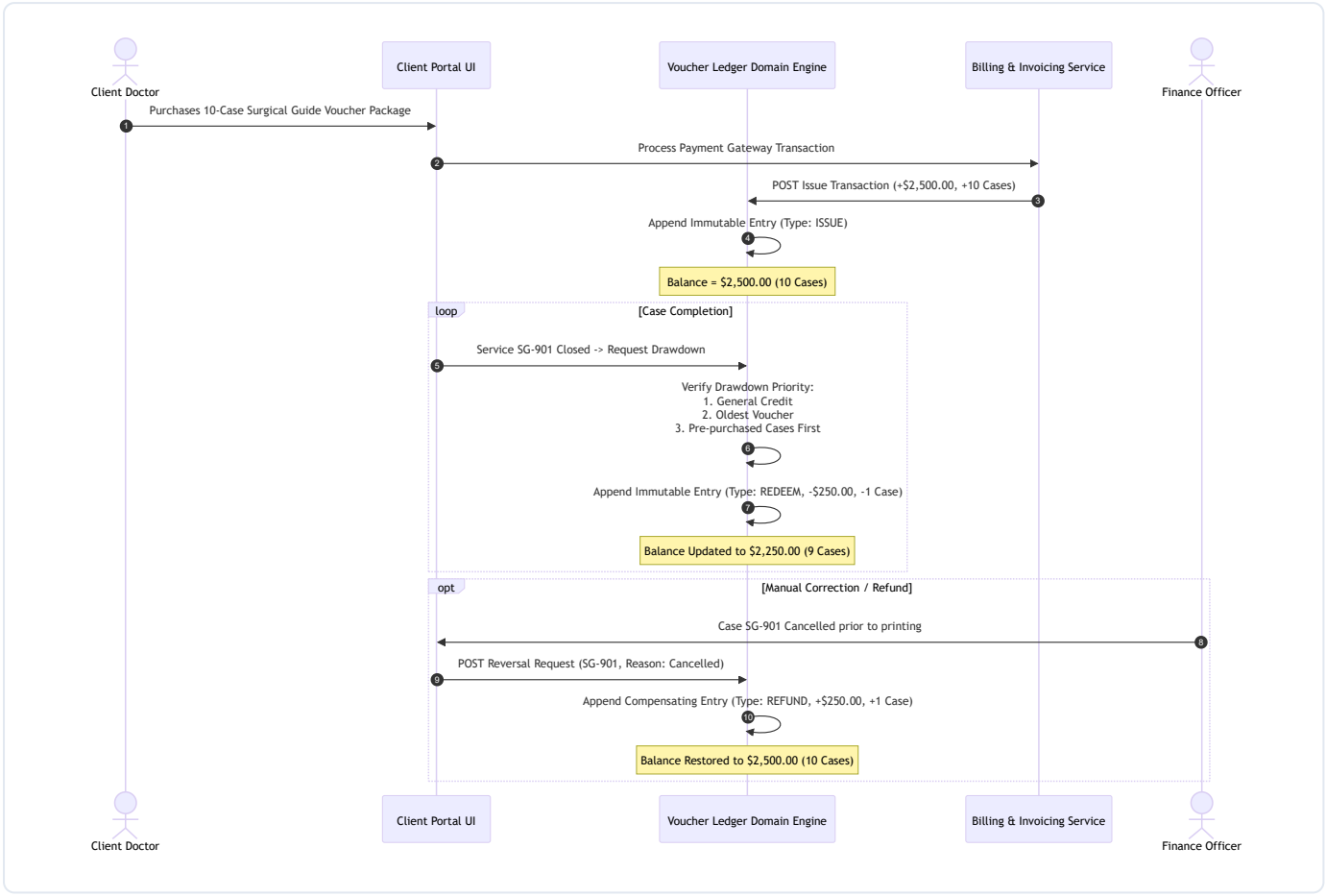
This sequence demonstrates the full flow from doctor submission to courier dispatch, highlighting the Cairo-Boston separation and the critical approval gate:


Dentist (Doctor)
Frontend UI
Cairo Operations (CAD)
Boston Lab (Physical)
Cairo Planning Team
Customer Support
Finance System
Submits Case, Charts Teeth & Uploads CBCT
Dispatches 3D Conversion Order
Converts DICOM & Maps Mandibular Canal
par
[Physical Courier Intake (Boston)]
Sends Stone Impression / Physical Model
Scans Barcode, Pours Model & Laser Scans
Uploads Optical Model STL & Bins Physical Package
[Digital Registration (Cairo)]
Registers Optical STL onto CBCT
Submits Registration Verdict (Option 1: Clean)
Unlocks Treatment Plan Order
Positions Virtual Implants in CoDiagnostiX
Sets Plan Status to "Ready" (Code 13)
Schedules & Conducts Fine-Tuning Video Call
critical
[Mandatory Approval Gate (No Manufacturing Before Doctor Sign-off)]
Clicks "Approve Treatment Plan" (or Confirms Email)
Validates Cryptographic Approval Token
Unlocks Sleeve Design (Code 38)
Positions Titanium Sleeves & Prints Instruction Sheet
Hands off to Planning QC (Code 47)
Models Guide CAD Body (Code 39)
Uploads Sliced Build File
Routes to Formlabs/Carbon 3D Printer (Code 42)
3D Prints Guide, IPA Wash & UV Curing Oven
Seats Titanium Sleeves & Tests on Fitting Surface (FS)
Packages Guide, Protocol & Scans Outbound Loaner Kit
Confirms Shipment with Tracking Number (Code 40)
Enters Daily Charging Cycle (Closed + Uncharged)
Deducts Voucher Credit (or Charges Credit Card)
Dentist (Doctor)
Frontend UI
Cairo Operations (CAD)
Boston Lab (Physical)
Cairo Planning Team
Customer Support
Finance System

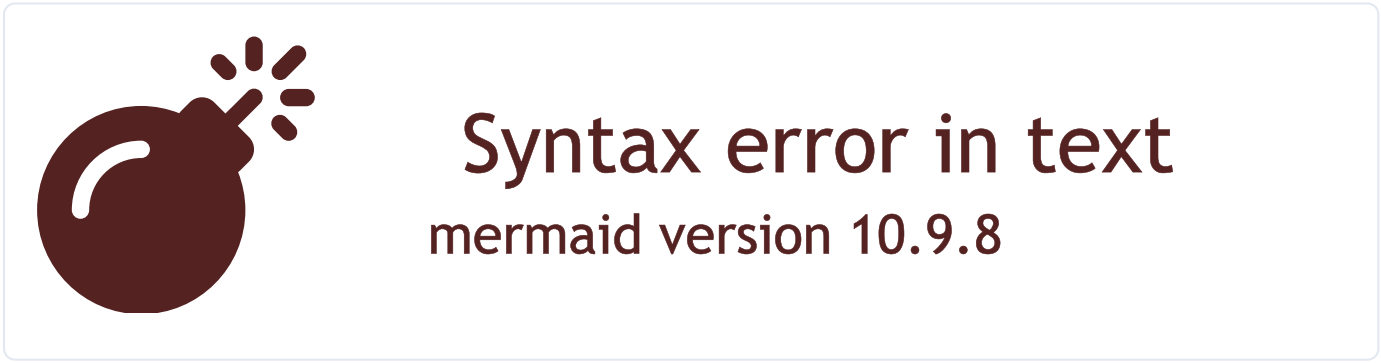
6.2 Registration Options 1–4 Decision Sequence



6.3 Immutable Voucher Ledger Transaction Sequence



6.4 Authentication, PKCE, Token Refresh & Tenant Isolation Sequence

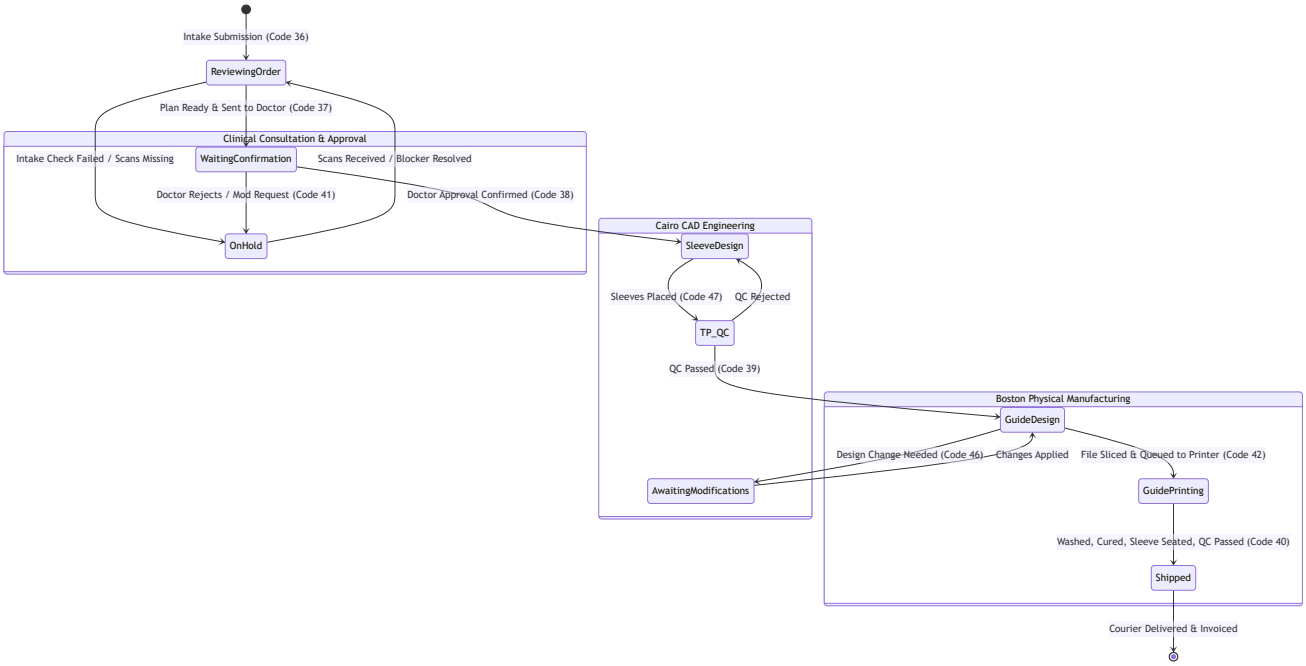


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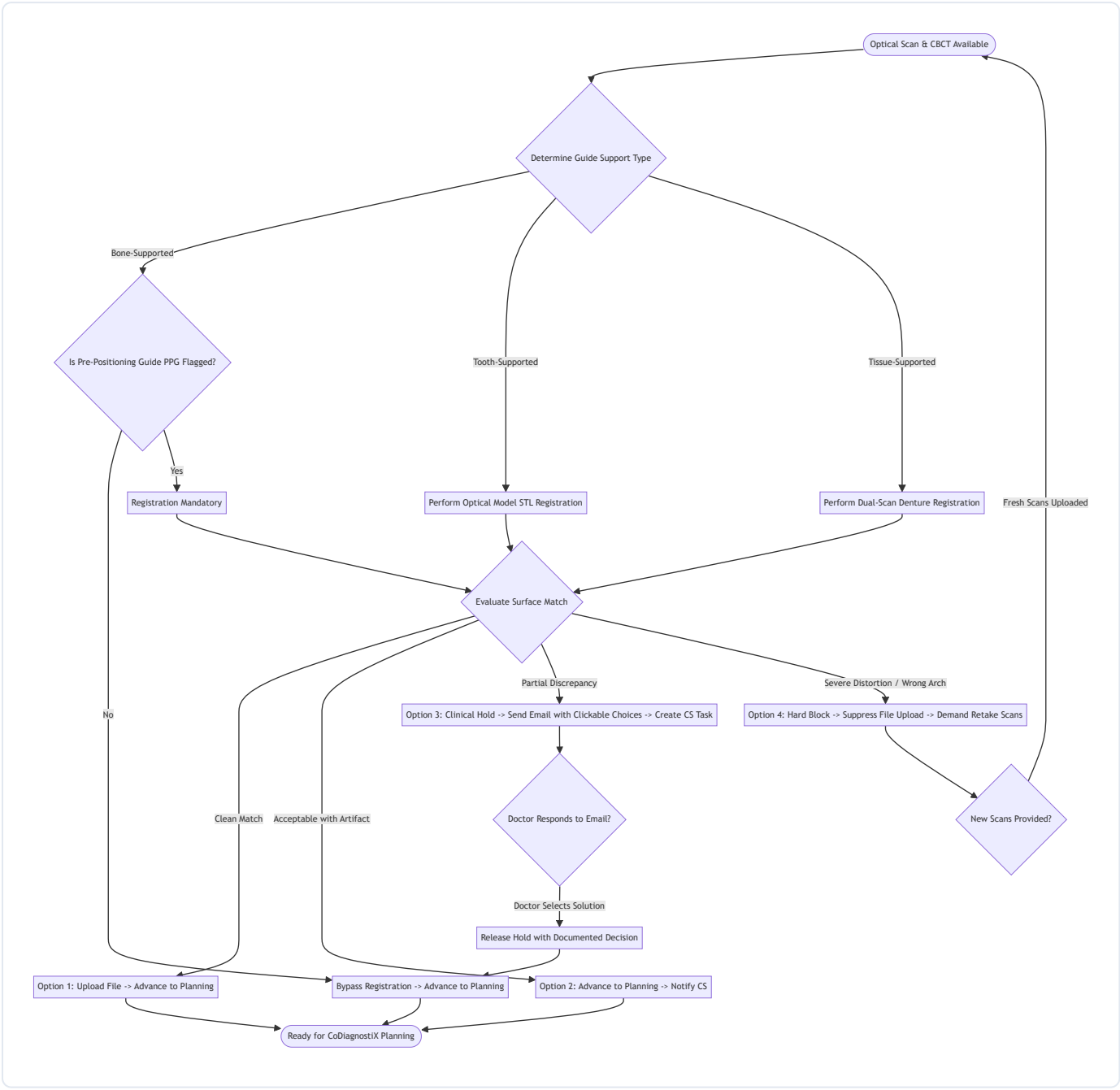
7. Comprehensive UML State Machine & Activity Diagrams

7.1 Surgical Guide Service-Line State Machine (`stateDiagram-v2`)

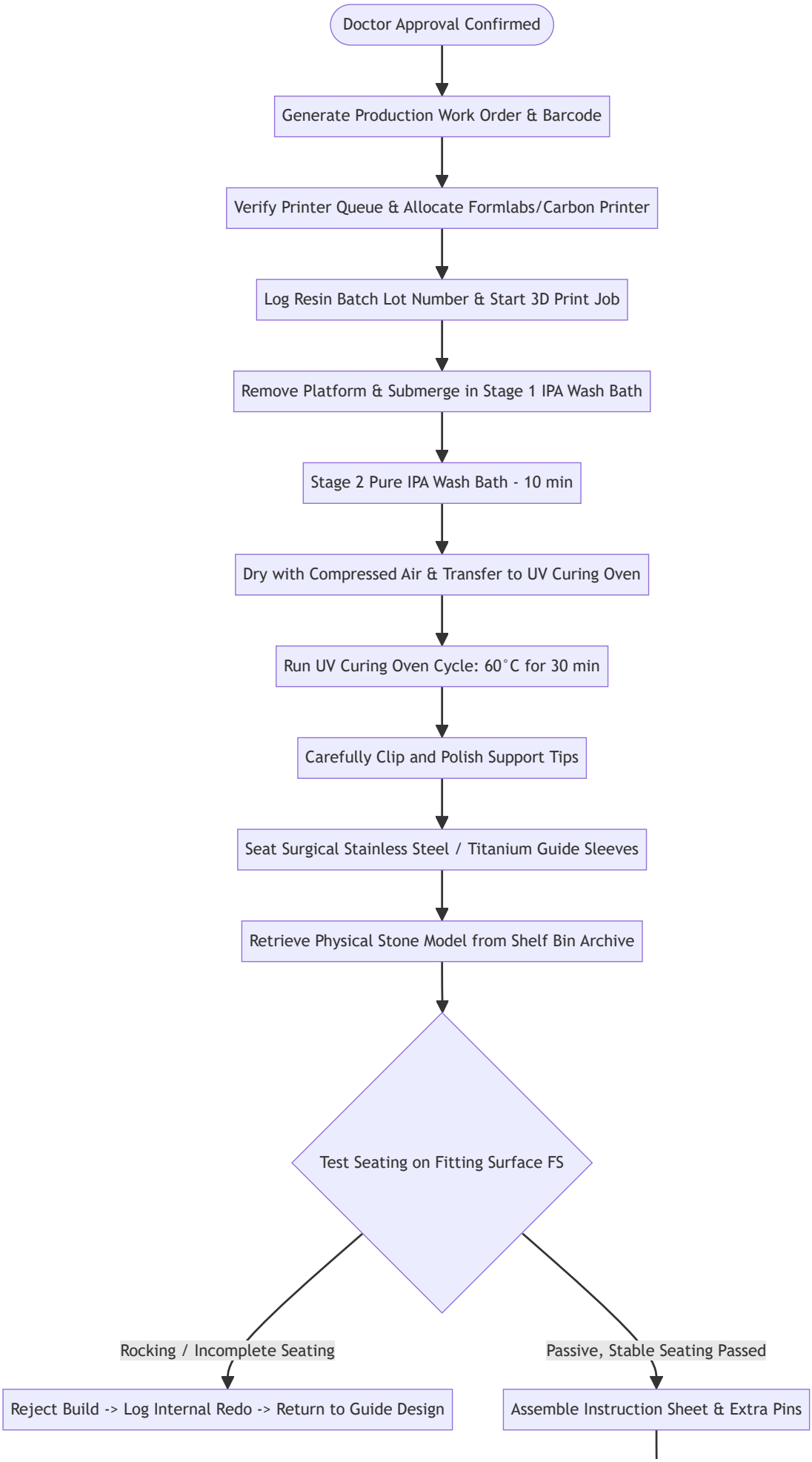
State integers reflect the exact 3DDX product vocabulary:

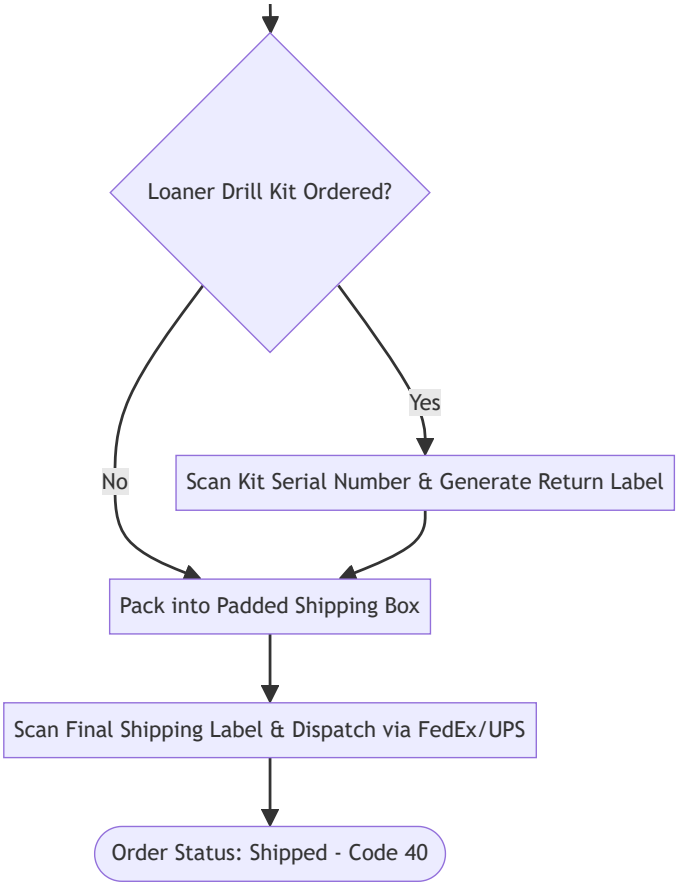

Intake Submission (Code 36)
ReviewingOrder
Plan Ready & Sent to Doctor (Code 37)
Clinical Consultation & Approval
Intake Check Failed / Scans Missing
WaitingConfirmation
Scans Received / Blocker Resolved
Doctor Rejects / Mod Request (Code 41)
Doctor Approval Confirmed (Code 38)
OnHold
Cairo CAD Engineering
SleeveDesign
Sleeves Placed (Code 47)
QC Rejected
TP_QC
QC Passed (Code 39)
AwaitingModifications
Boston Physical Manufacturing
GuideDesign
Design Change Needed (Code 46)
Changes Applied
File Sliced & Queued to Printer (Code 42)
GuidePrinting
Washed, Cured, Sleeve Seated, QC Passed (Code 40)
Shipped
Courier Delivered & Invoiced

7.2 Registration Verdict Activity Diagram



7.3 Boston Regulated Manufacturing Activity Flow


Doctor Approval Confirmed
Generate Production Work Order & Barcode
Verify Printer Queue & Allocate Formlabs/Carbon Printer
Log Resin Batch Lot Number & Start 3D Print Job
Remove Platform & Submerge in Stage 1 IPA Wash Bath
Stage 2 Pure IPA Wash Bath - 10 min
Dry with Compressed Air & Transfer to UV Curing Oven
Run UV Curing Oven Cycle: 60°C for 30 min
Carefully Clip and Polish Support Tips
Seat Surgical Stainless Steel / Titanium Guide Sleeves
Retrieve Physical Stone Model from Shelf Bin Archive
Test Seating on Fitting Surface FS
Rocking / Incomplete Seating
Passive, Stable Seating Passed
Reject Build -> Log Internal Redo -> Return to Guide Design
Assemble Instruction Sheet & Extra Pins



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8. Authentication, Tenant Isolation & Privacy Boundaries

8.1 Tenant Context Resolution

- The application enforces strict multi-tenant boundaries. Every outgoing API request carries an authenticated `X-Tenant-ID` header.
- For Dental Support Organizations (DSOs), switching clinics triggers an automatic eviction of the React Query cache to prevent cross-tenant data bleed.

8.2 Anonymized Patient Privacy Architecture

In compliance with Section 13.9 of the 3DDX Business Guide:

1. `clinical.patients` contains only `patient_code` and `client_id`.
2. Real patient identities (name, phone, DOB) are isolated in `clinical.patient_identities`.
3. The frontend never displays patient names in standard lists. Clicking "Show Identity" prompts the user for a clinical purpose, dispatching a logged audit event before returning the identity token.

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9. Docker and Containerization Strategy

9.1 Multi-Stage Production Dockerfile Design

```
# Stage 1: Build Environment
FROM node:22-alpine AS builder
WORKDIR /app
COPY package.json package-lock.json ./
RUN npm ci --prefer-offline --no-audit
COPY . .
RUN npm run build

# Stage 2: Hardened Nginx Runtime
FROM nginx:alpine AS runner
WORKDIR /usr/share/nginx/html
RUN rm -rf ./*
COPY --from=builder /app/dist .
COPY deploy/nginx.conf /etc/nginx/conf.d/default.conf

USER nginx
EXPOSE 8080
HEALTHCHECK --interval=30s --timeout=3s --retries=3 \
  CMD wget --quiet --tries=1 --spider http://localhost:8080/health || exit 1
CMD ["nginx", "-g", "daemon off;"]
```

9.2 Custom Nginx Routing & Security Headers (`deploy/nginx.conf`)

- **SPA Fallback Routing:** `try_files $uri $uri/ /index.html;` ensures React Router deep linking functions without 404 errors.
- **Security Headers:** CSP, HSTS, X-Frame-Options, X-Content-Type-Options, Referrer-Policy.

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10. Automated Testing Strategy

Test Level	Target Framework	Scope & Coverage Targets	Execution Trigger
Unit Tests	Vitest	Pure domain functions, pricing calculators, status decoders, voucher drawdowns. Target: > 90% logic coverage.	Pre-commit / CI PR
Component Tests	Vitest + React Testing Library	UI components, form validations, accessibility keyboard traps, empty/error state handling.	Pre-commit / CI PR
Contract / API Tests	MSW + Vitest	Verifies that frontend API adapters correctly parse backend JSON payloads against OpenAPI schemas.	CI PR
End-to-End (E2E)	Playwright (Future Phase)	Critical user journeys: Case creation -> 3D Odontogram -> Approval Gate -> Billing dispatch.	Nightly / Pre-release

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11. Security, Compliance & Observability

1. **Zero Client Secrets:** All authentication exchanges use OAuth PKCE or server-side session cookies. No API private keys exist in client code.
2. **Cross-Site Scripting (XSS) Mitigation:** React's built-in JSX escaping prevents HTML injection. Raw SVG/HTML rendering is strictly prohibited without DOMPurify sanitization.
3. **Structured Telemetry:** Production errors caught by React Error Boundaries are transmitted with correlation IDs, route context, and anonymized user IDs.

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12. Architectural Decisions Requiring Human Approval

The following critical decisions require explicit sign-off from the Technical Lead and Product Owner before code execution:

1. **Selection of Unit Testing Engine:** Formal adoption of Vitest over Jest for alignment with Vite.
2. **Adoption of TanStack Query:** Replacing `ReactiveStore` with React Query for server-state synchronization.
3. **OpenAPI Specification Freeze:** Confirmation of the final OpenAPI 3.0.3 contract from the backend engineering team.
4. **Approval of FMP/GFMP Status Sequences:** Finalizing the open business question regarding whether FMP/GFMP are independent services, order forms, or bundled workflows.

Authored under Task T-000 governance. Canonical project-level software design specification for `newcp-ui-option-a`.